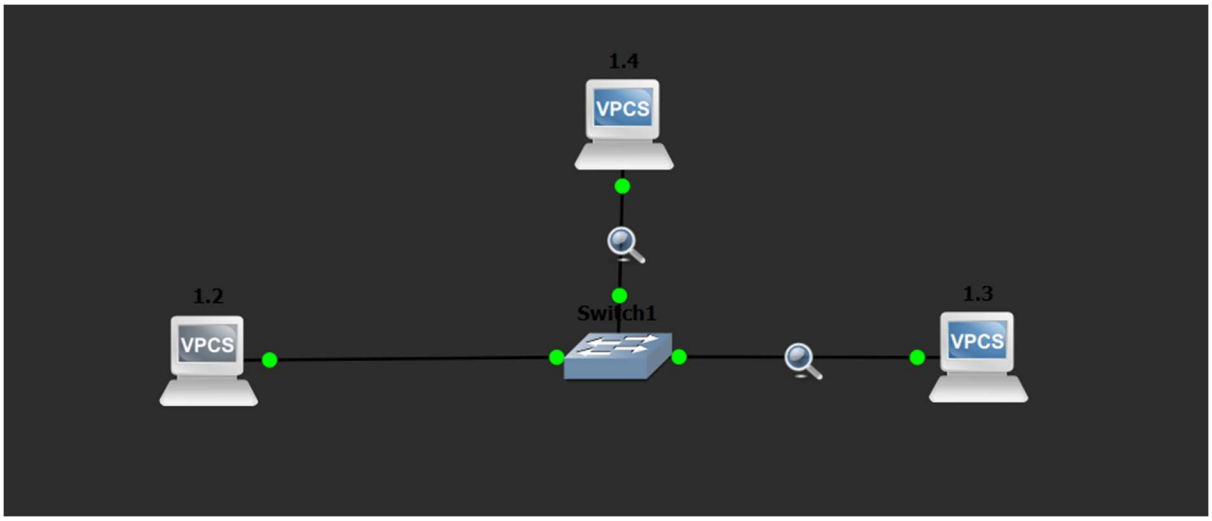
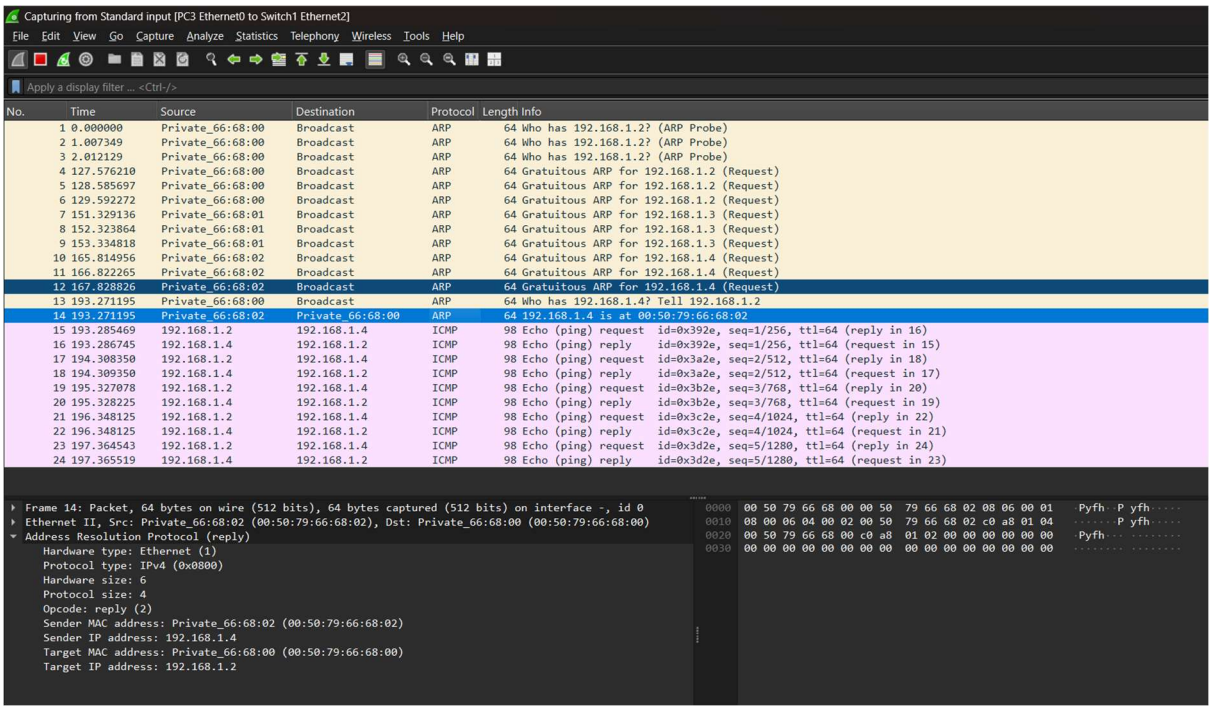
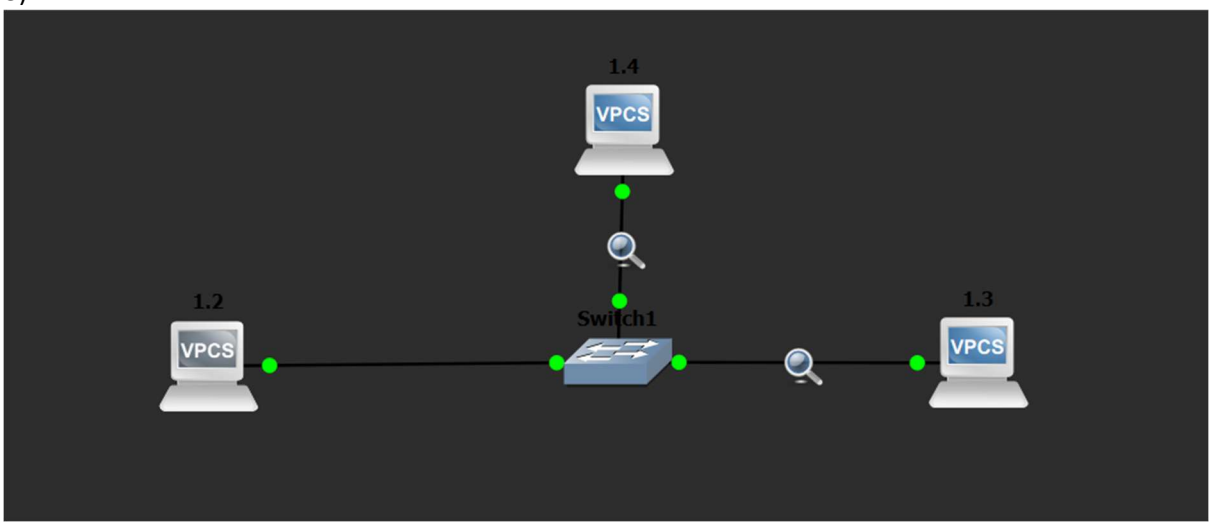
1) ARP ANALYSIS

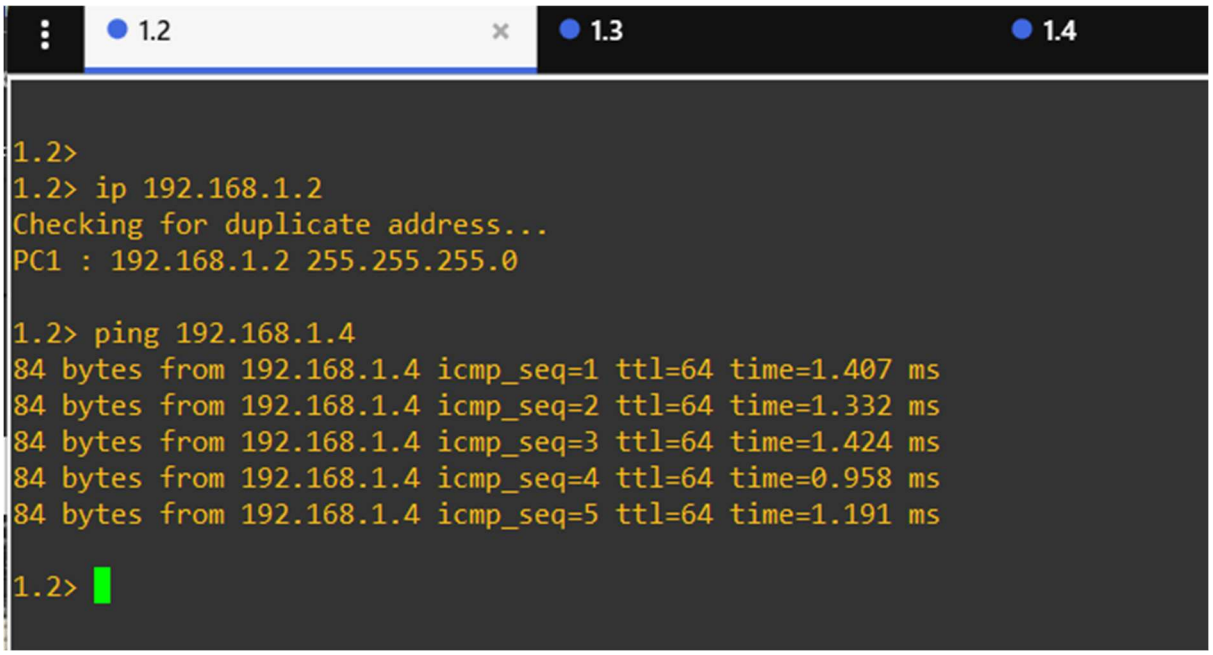


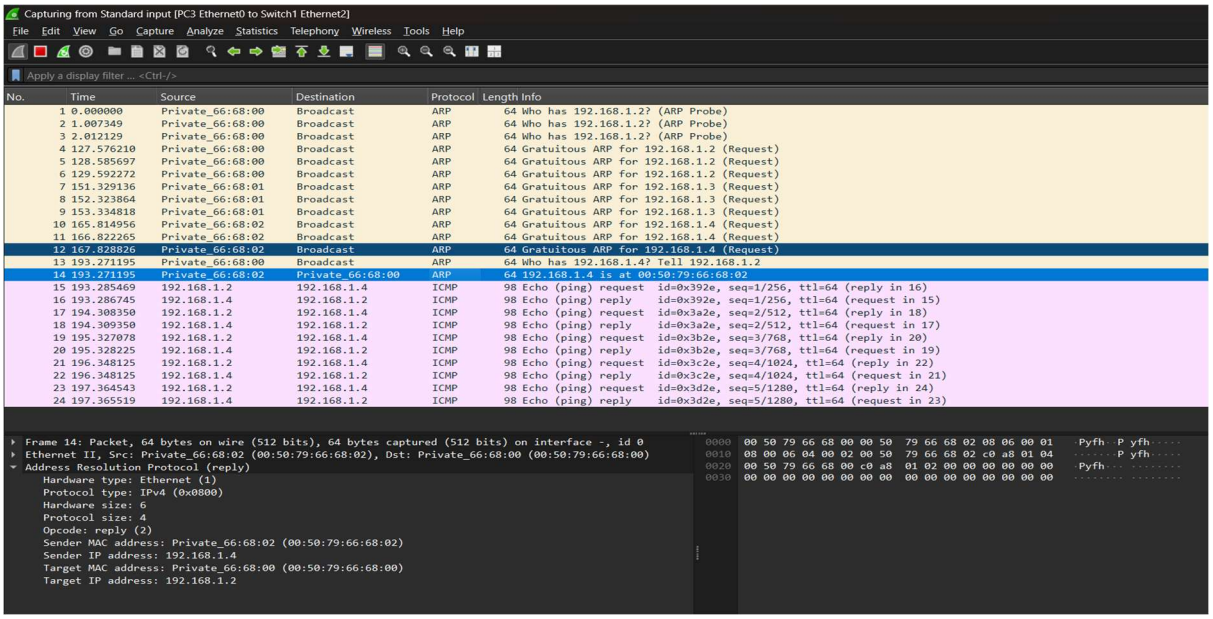


2)

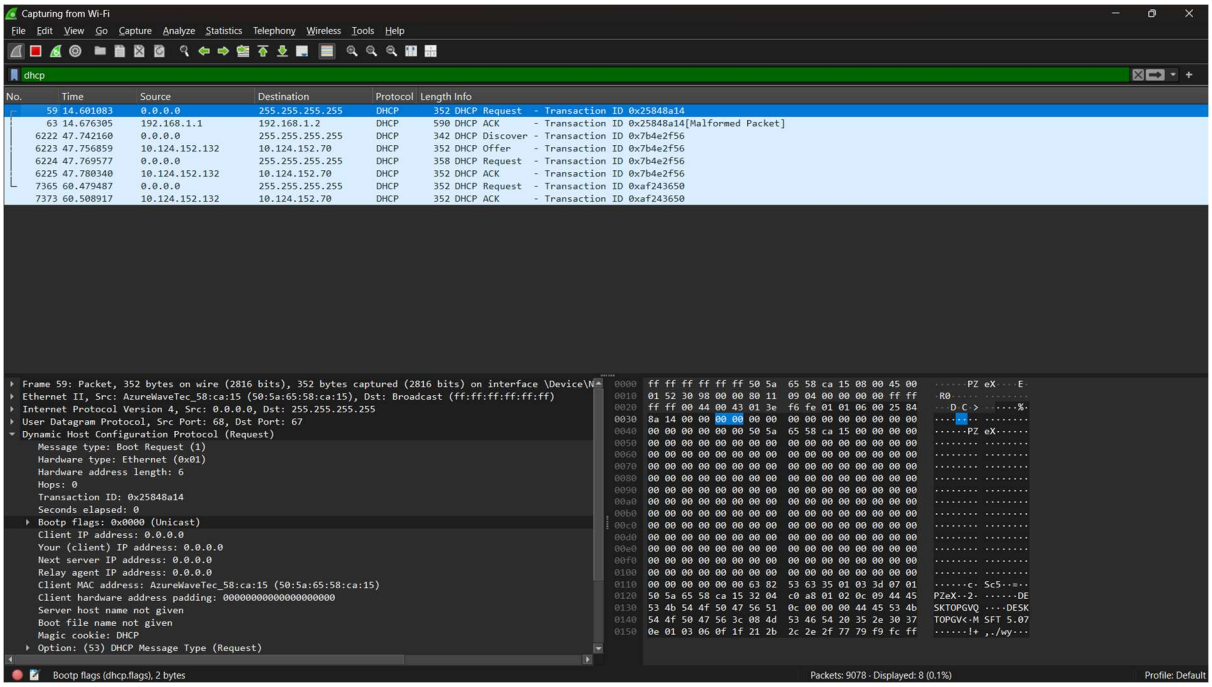
3)


1.4
VPCS
1.2
VPCS
Switch1
1.3
VPCS


1.2
1.3
1.4
1.2>
1.2> ip 192.168.1.2
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0
1.2> ping 192.168.1.4
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=1.407 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=1.332 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=1.424 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=0.958 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=1.191 ms
1.2>


Capturing from Standard input [PC3 Ethernet0 to Switch1 Ethernet2]
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 Private_66:68:00 Broadcast ARP 64 Who has 192.168.1.2? (ARP Probe)
2 1.007349 Private_66:68:00 Broadcast ARP 64 Who has 192.168.1.2? (ARP Probe)
3 2.012129 Private_66:68:00 Broadcast ARP 64 Who has 192.168.1.2? (ARP Probe)
4 127.576210 Private_66:68:00 Broadcast ARP 64 Gratuitous ARP for 192.168.1.2 (Request)
5 128.585697 Private_66:68:00 Broadcast ARP 64 Gratuitous ARP for 192.168.1.2 (Request)
6 129.592272 Private_66:68:00 Broadcast ARP 64 Gratuitous ARP for 192.168.1.2 (Request)
7 151.329136 Private_66:68:01 Broadcast ARP 64 Gratuitous ARP for 192.168.1.3 (Request)
8 152.323864 Private_66:68:01 Broadcast ARP 64 Gratuitous ARP for 192.168.1.3 (Request)
9 153.334818 Private_66:68:01 Broadcast ARP 64 Gratuitous ARP for 192.168.1.3 (Request)
10 165.814956 Private_66:68:02 Broadcast ARP 64 Gratuitous ARP for 192.168.1.4 (Request)
11 166.822265 Private_66:68:02 Broadcast ARP 64 Gratuitous ARP for 192.168.1.4 (Request)
12 167.828826 Private_66:68:02 Broadcast ARP 64 Gratuitous ARP for 192.168.1.4 (Request)
13 193.271195 Private_66:68:00 Broadcast ARP 64 Who has 192.168.1.4? Tell 192.168.1.2
14 193.271195 Private_66:68:02 Private_66:68:00 ARP 64 192.168.1.4 is at 00:50:79:66:68:02
15 193.285469 192.168.1.2 192.168.1.4 ICMP 98 Echo (ping) request id=0x392e, seq=1/256, ttl=64 (reply in 16)
16 193.286745 192.168.1.4 192.168.1.2 ICMP 98 Echo (ping) reply id=0x392e, seq=1/256, ttl=64 (request in 15)
17 194.308350 192.168.1.2 192.168.1.4 ICMP 98 Echo (ping) request id=0x3a2e, seq=2/512, ttl=64 (reply in 18)
18 194.309350 192.168.1.4 192.168.1.2 ICMP 98 Echo (ping) reply id=0x3a2e, seq=2/512, ttl=64 (request in 17)
19 195.327078 192.168.1.2 192.168.1.4 ICMP 98 Echo (ping) request id=0x3b2e, seq=3/768, ttl=64 (reply in 20)
20 195.328225 192.168.1.4 192.168.1.2 ICMP 98 Echo (ping) reply id=0x3b2e, seq=3/768, ttl=64 (request in 19)
21 196.348125 192.168.1.2 192.168.1.4 ICMP 98 Echo (ping) request id=0x3c2e, seq=4/1024, ttl=64 (reply in 22)
22 196.348125 192.168.1.4 192.168.1.2 ICMP 98 Echo (ping) reply id=0x3c2e, seq=4/1024, ttl=64 (request in 21)
23 197.364543 192.168.1.2 192.168.1.4 ICMP 98 Echo (ping) request id=0x3d2e, seq=5/1280, ttl=64 (reply in 24)
24 197.365519 192.168.1.4 192.168.1.2 ICMP 98 Echo (ping) reply id=0x3d2e, seq=5/1280, ttl=64 (request in 23)
Frame 14: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
Ethernet II, Src: Private_66:68:02 (00:50:79:66:68:02), Dst: Private_66:68:00 (00:50:79:66:68:00)
Address Resolution Protocol (reply)
Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: reply (2)
Sender MAC address: Private_66:68:02 (00:50:79:66:68:02)
Sender IP address: 192.168.1.4
Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
Target IP address: 192.168.1.2
0000 00 50 79 66 68 00 00 50 79 66 68 02 08 06 00 01 Pyfh P yfh
0010 08 00 06 04 00 02 00 50 79 66 68 02 c0 a8 01 04 P yfh
0020 00 50 79 66 68 00 c0 a8 01 02 00 00 00 00 00 00 Pyfh
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

4)



5)

Step 1: Determine New Subnet Mask

- Given: /24
- Need: **4 subnets**
- Formula:

$$2^n = 4 \Rightarrow n = 2$$

Borrow 2 bits

New Subnet Mask:

/26 → 255.255.255.192

Step 2: Subnet Details

Each subnet has:

- Total IPs = 64
- Usable hosts = 62

Step 3: Subnet Ranges

Subnet	Network Address	Usable Range	Broadcast
1	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
2	192.168.1.64/26	192.168.1.65 – 192.168.1.126	192.168.1.127

Subnet	Network Address	Usable Range	Broadcast
3	192.168.1.128/26	192.168.1.129 – 192.168.1.190	192.168.1.191
4	192.168.1.192/26	192.168.1.193 – 192.168.1.254	192.168.1.255

Step 4: Packet Tracer Assignment (Example)

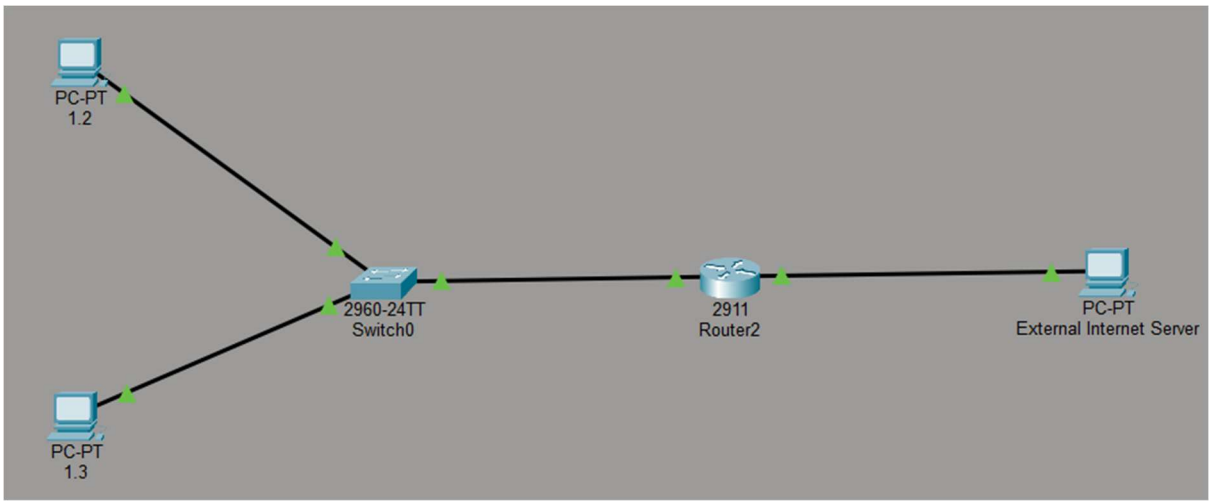
- Subnet 1:
  - PC1 → 192.168.1.1
  - PC2 → 192.168.1.2
- Subnet 2:
  - PC3 → 192.168.1.65

Devices in same subnet → **Ping works**
Different subnet → **Need router**

6)

IP Address	Class	Default Subnet Mask
10.1.1.1	Class A	255.0.0.0 (/8)
172.16.5.10	Class B	255.255.0.0 (/16)
192.168.1.5	Class C	255.255.255.0 (/24)

7) PRIVATE IP+NAT to access a server via Internet



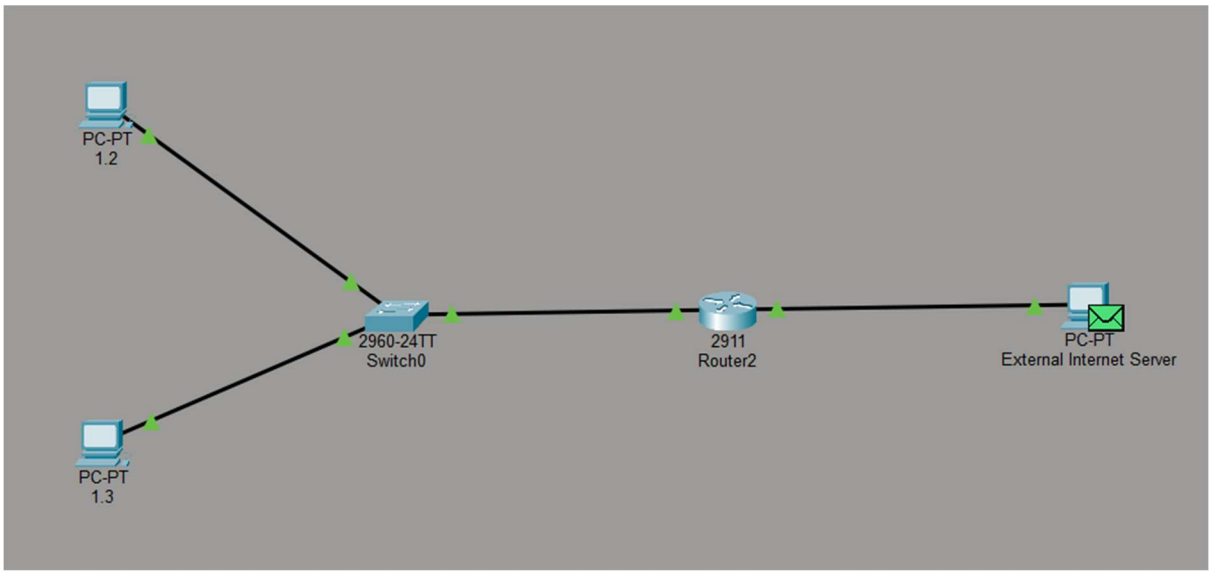
Router2

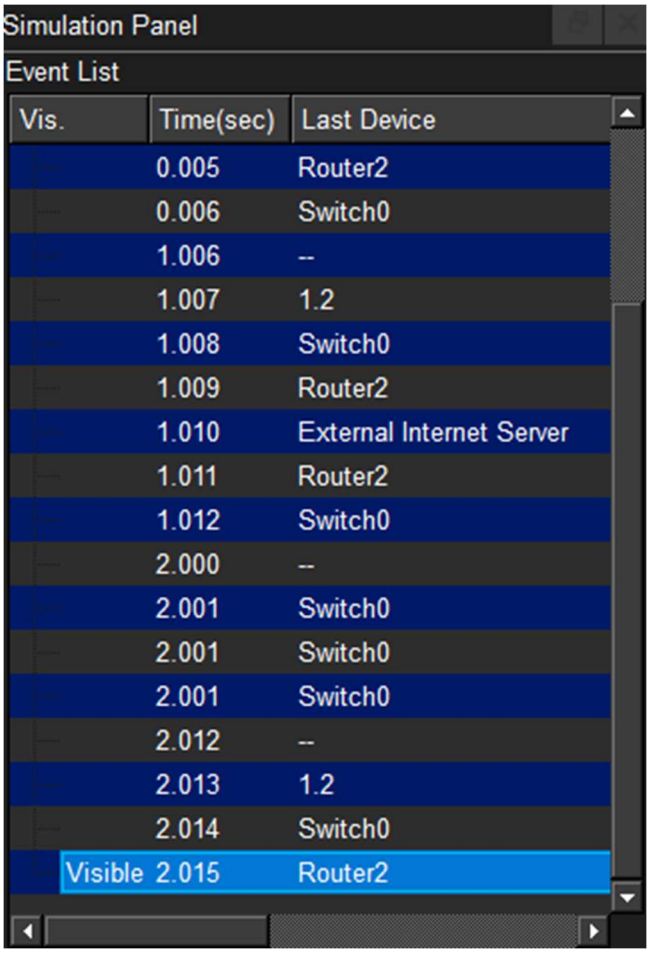
Physical | Config | CLI | Attributes

```
Router>
Router>
Router>
Router>
Router>
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#enable
                  ^
% Invalid input detected at '^' marker.

Router(config-if)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#
```

Copy | Paste

Top





Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.005	Router2
	0.006	Switch0
	1.006	--
	1.007	1.2
	1.008	Switch0
	1.009	Router2
	1.010	External Internet Server
	1.011	Router2
	1.012	Switch0
	2.000	--
	2.001	Switch0
	2.001	Switch0
	2.001	Switch0
	2.012	--
	2.013	1.2
	2.014	Switch0
Visible	2.015	Router2

Before NAT, the source IP address is 192.168.1.2 (private IP).
After NAT, the source IP address becomes 10.0.0.1 (router's public interface IP).